

Enforcing Kantian Equilibria:

From Nash's Prisoner's Dilemma to Learning Cooperation

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■ The Core Idea — in plain language

When people act purely in their own interest, they often end up worse off than if they had worked together. This paper asks: can we design a simple rule — a multiplier (λ) and a penalty (δ) — that makes self-interested players voluntarily cooperate? We prove the answer is yes, and we show how cooperation can be learned over time and applied to global challenges like climate change and UN Security Council voting.

1 The Problem: Why Do People Not Cooperate?

Imagine two countries negotiating climate commitments. If each country only thinks about itself, it will try to "free ride" — let the other country make expensive cuts while benefiting from the cleaner atmosphere at no cost. The tragic result: both defect, total emissions barely fall, and everyone loses. This is the famous **Prisoner's Dilemma**.

John Nash proved that when every player acts selfishly, the outcome is a **Nash Equilibrium**: individually rational, but collectively disastrous. In a Prisoner's Dilemma, the Nash Equilibrium is mutual defection — bad for everyone, yet no single player can improve by deviating alone.

■ The Prisoner's Dilemma at a Glance

Two players each choose: Cooperate (C) or Defect (D).

- If both cooperate → each gets reward R.
- If one defects while the other cooperates → defector gets T (temptation), cooperator gets S (sucker).
- If both defect → each gets P (punishment).

Because $T > R > P > S$, defecting always looks tempting — but mutual defection (P,P) is worse than cooperation (R,R).

1.2 The Kantian Solution

The philosopher Immanuel Kant proposed a moral rule: *"Act only according to that maxim which you can at the same time will to become a universal law."* Translated into game theory by economist John Roemer, the **Kantian Equilibrium** asks each player a different question.

Instead of asking "What is my best response, taking others' actions as fixed?" (Nash), the Kantian player asks: **"If everyone scaled their effort by the same factor λ simultaneously, would I want to propose $\lambda \neq 1$?"** At a Kantian Equilibrium, the answer is no — no player benefits from any proportional deviation.

■ Key Insight

Nash asks: "What should I do, given others?" → leads to free-riding.

Kant asks: "What should everyone do together?" → leads to cooperation.

The Kantian Equilibrium Pareto-dominates the Nash Equilibrium: everyone is strictly better off.

2 What Does the Literature Say?

This paper builds on several streams of research spanning game theory, ethics, biology, and international relations:

- **Game Theory:** Nash (1950, 1951) established non-cooperative equilibrium; Axelrod (1984) showed cooperative strategies can thrive in repeated games.
- **Kantian Equilibrium:** Roemer (2010, 2019) formalised the concept and proved it achieves efficient outcomes in public-good settings.
- **Mechanism Design:** Fehr & Gächter (2000) demonstrated experimentally that punishment mechanisms substantially expand cooperation.

- **Learning in Games:** Kalai & Lehrer (1993) proved Bayesian learners converge to Nash equilibria; Honkapohja applied Bayesian updating to macroeconomic dynamics.
- **Evolutionary Dynamics:** Weibull (1995) provides the canonical replicator dynamic; cooperation is evolutionarily stable under sufficient enforcement.
- **International Cooperation:** Barrett (2003) on self-enforcing environmental agreements; Nordhaus (2015) on climate clubs; Voeten (2005) on the UN Security Council.

3 How to Enforce Kantian Equilibria: The λ - δ Mechanism

Our core contribution is a two-parameter enforcement mechanism that moves players from Nash non-cooperation to Kantian cooperation.

3a Scaling Factor $\lambda > 1$

An institution (a treaty, a regulator, an international body) applies a multiplier $\lambda > 1$ to all players' submitted efforts. Because $\lambda > 1$, the effective total production is amplified, and the benefits of cooperation exceed what Nash play delivers. Players are incentivised to raise their effort toward the Kantian level.

3b Equiproportionality Penalty $\delta > 0$

If two players exert different efforts — if one free-rides while the other works harder — both bear a penalty proportional to the difference: $\delta \times |e_i - e_j|$. This enforces symmetric, fair burden-sharing. Any player who contributes asymmetrically faces an immediate cost.

■ ■ The Mechanism in One Sentence

λ raises the reward for collective effort; δ punishes unequal sharing — together they make mutual cooperation the uniquely rational strategy.

Theorem 1 (Existence): For any target effort between Nash and socially optimal, a pair (λ^*, δ^*) exists with $\lambda^* > 1$ and $\delta^* > 0$ such that the unique equilibrium of the modified game is exactly the target. The welfare at this equilibrium strictly exceeds the Nash welfare.

4 How Cooperation is Learned Over Time

4.1 Bayesian Learning

Players who start with uncertain beliefs and update them via Bayes' Rule after each round converge to the enforcement equilibrium with probability one (Theorem 2). The mechanism eliminates off-equilibrium attractors, so learning always leads to cooperation.

4.2 Evolutionary Dynamics (Weibull)

In a population where strategies spread by imitation, the replicator dynamic shows: without enforcement, cooperators die out; with $\delta > \lambda(T-R)$, the entire population converges to full cooperation regardless of the starting point (Proposition 4).

4.3 From Lizards to Humans

Commitment to symmetric effort is an ancient strategy. Male side-blotched lizards use colour signals to enforce cooperative territorial sharing — a biological δ -penalty. Anthropological evidence (Bowles 2006) shows between-group conflict sustained within-group altruism in early humans. Neuroeconomic experiments reveal that the human brain's insula and dorsal striatum encode norm violations and punishment — a neural substrate for cooperative enforcement.

5 Applications to International Organisations

5.1 The Global Climate Game

Climate change is the most consequential global Prisoner's Dilemma. Each country benefits from others' reductions but bears the full cost of its own abatement. Without enforcement, every country free-rides.

5.1.1 Game with all 195 UNFCCC Countries

With 195 parties, logarithmic global benefits, and quadratic individual costs, Nash abatement totals 18.3 units. Under Kantian enforcement ($\lambda^* = 1.71$, $\delta^* = 0.42$) it rises to 31.2 — a 70% absolute increase, closing 90% of the Nash–optimum gap.

5.1.2 Game with the Six Largest Polluters + Rest of World

China (28.1%), USA (13.7%), India (10.1%), Russia (5.0%), Japan (3.0%), and Germany (2.0%) account for 68% of global emissions. Modelling them as one bloc vs. the Rest of the World yields a clean two-player game: Kantian enforcement generates a Pareto improvement for both blocs when λ^* exceeds the threshold of Proposition 3.

5.2 UN Security Council: P5 Veto Power

The UNSC has 5 permanent members (P5) with veto power and 10 non-permanent members (NP10) without. A resolution requires 9 yes-votes and no P5 veto. Under Nash strategic voting, resolutions pass 42% of the time. Under the Kantian cooperative norm (enforced by coalition burden-sharing λ and General Assembly reputational penalties δ), passage rises to 67%.

6 Results at a Glance

33%	Welfare gain Kantian vs Nash 1.71	Optimal λ^* scaling factor 0.42	Optimal δ^* penalty parameter 195
Countries in climate game			
Metric	Nash (No Cooperation)	Kantian (Enforced)	Social Optimum
Total abatement	18.3 units	31.2 units	34.6 units
Global welfare	142.1	189.7	195.3

Metric	Nash (No Cooperation)	Kantian (Enforced)	Social Optimum
Scaling factor λ	1.00 (none)	1.71	—
Penalty δ	0 (none)	0.42	—
Welfare gain	Baseline	+33.5%	+37.4%
UNSC passage rate	42%	67%	71%

The numbers confirm: Kantian enforcement delivers welfare gains of ~33% relative to Nash, without requiring altruism or centralised authority. Players remain self-interested; the mechanism changes what self-interest recommends.

7 Discussion

Real-World Parallels

In climate policy, $\lambda > 1$ corresponds to the Paris Agreement's ratchet clause — NDC commitments amplified over successive review rounds. $\delta > 0$ corresponds to Border Carbon Adjustments that reward symmetric effort. In the UNSC, λ maps to NATO-style burden-sharing pledges; δ maps to General Assembly reputational sanctions for strategic vetoes.

Limitations

- **Heterogeneity:** Asymmetric enforcement for players with different cost structures is left for future work.
- **Private Information:** A mechanism-design extension with incentive-compatible type revelation is a natural next step.
- **Renegotiation:** The threat of imposing δ must be credible ex post; renegotiation-proofness requires additional structure.
- **Empirical Calibration:** Structural estimation using observed NDC submissions and UNSC voting data remains to be done.

8 Conclusion

We have shown that Kantian equilibria can be enforced through a simple two-parameter mechanism: a scaling factor $\lambda > 1$ that amplifies collective rewards, and a symmetry penalty $\delta > 0$ that deters free-riding. The mechanism works under Bayesian learning, evolutionary dynamics, and across the full spectrum from lizard territorial signalling to UN Security Council reform.

Applied to global climate governance, it delivers a 33% welfare gain over Nash and closes 90% of the cooperation shortfall. Applied to the UNSC, it raises resolution passage rates by 25 percentage points. These are not marginal improvements — they represent the difference between a world that solves its hardest collective-action problems and one that does not.

■ The Bottom Line

Smart institutional design — a multiplier and a penalty — can transform self-interest into cooperation. The tools already exist in embryonic form in the Paris Agreement and the UN Charter. This paper provides the mathematical proof that they can work, and the precise conditions under which they will.

9 Key References

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Full bibliography with DOIs in the technical paper: [kant_equilibrium.pdf](#)

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